

 Technical Specification: Homura_Recon
Project: Aegis Blue-Team Framework
Component: Network Surface Reconnaissance Module
Classification: Defensive Security Tooling
Version: 1.0.0

1. Executive Summary

Homura_Recon is a specialized network auditing tool developed for Blue Team operations. Unlike traditional scanners designed for offensive penetration testing, Homura_Recon is engineered for **Defensive Surface Mapping**. Its primary objective is to provide security administrators with an immediate and transparent view of the exposed attack surface of a target host or network segment.

2. Operational Objectives

The tool is designed to fulfill three critical defensive requirements:

- * **Attack Surface Reduction:** Identify unnecessary open ports that should be closed to minimize the entry points for attackers.
- * **Service Verification:** Ensure that only authorized services (e.g., SSH, HTTPS) are running and exposed.
- * **Baseline Auditing:** Establish a "known-good" state of network services to detect unauthorized service installations (shadow IT) or backdoor listeners.

3. Technical Architecture

3.1 Core Engine

The tool utilizes a **Multi-threaded Socket Architecture**, enabling high-speed scanning without the overhead of heavy external dependencies. By implementing a concurrent scanning model, it minimizes the time-to-result while maintaining low system resource consumption.

3.2 Scanning Logic

- * **TCP Connect Scanning:** Employs a standard TCP three-way handshake to verify port availability with high reliability.
- * **Service Identification:** Performs basic banner grabbing and port-to-service mapping to identify the nature of the exposed application.
- * **Targeting:** Supports single-host auditing and targeted range scanning for subnet verification.

4. Defensive Application (Blue Teaming)

In a professional SOC (Security Operations Center) or incident response scenario, Homura_Recon is applied as follows:

1. **Post-Incident Audit:** After a suspected breach, it is used to check for newly opened ports that may indicate a persistence mechanism (Reverse Shells).
2. **Hardening Validation:** After applying firewall rules, it verifies that the ports are indeed unreachable from the outside.
3. **Compliance Checking:** Regularly audits servers to ensure they adhere to the organization's "Minimal Services" policy.

5. Conclusion

Homura_Recon provides the transparency necessary for effective perimeter defense. By shifting the focus from "finding vulnerabilities" to "managing exposure," it serves as a critical asset in the Aegis Framework's detection phase.

Generated by Aegis Framework Documentation Engine